

5. Audio coding standards

Audio coding · based on Spanias, Painter & Atti, Chapter 10

Key question: how do standard codecs combine transforms, psychoacoustics, quantization and bitstream syntax?

Learning goals

By the end of this topic, students should be able to compare major audio coding standards and their design choices.

- Distinguish MIDI control data from sampled audio formats.
- Explain how mono, stereo and surround formats change coding requirements.
- Summarize the evolution from MPEG-1/MP3 to AAC and MPEG-4 Audio.
- Identify the main components of MP3, AAC and Dolby Digital encoders.
- Explain why FLAC is lossless and how it differs from perceptual codecs.
- Use FFmpeg as a practical tool for transcoding and inspecting audio files.

Roadmap

1 · Representations

- MIDI format
- Mono, stereo, surround
- MPEG standards overview

2 · Perceptual standards

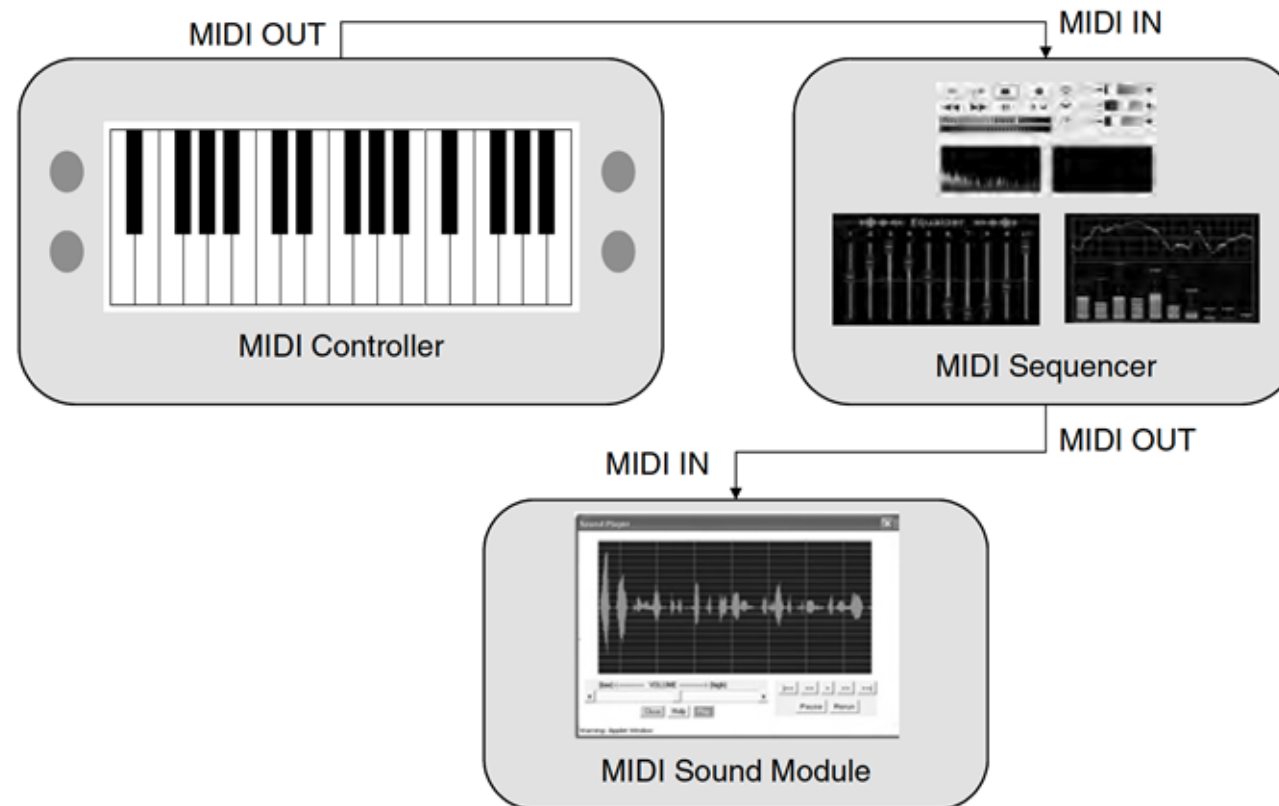
- MPEG-1 Layer 3 (MP3)
- MPEG-2 AAC
- MPEG-4 Audio
- Dolby Digital (AC-3)

3 · Lossless and tools

- FLAC
- FFmpeg
- Key takeaways
- References

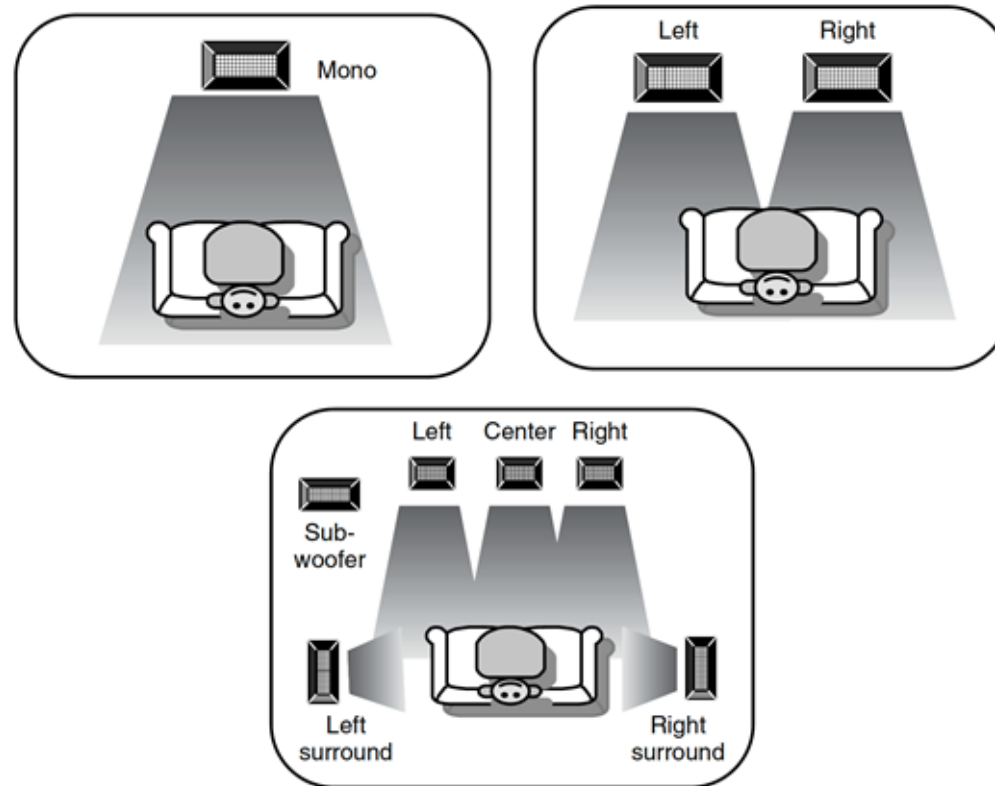
MIDI: symbolic control, not audio samples

MIDI transmits performance events and control messages between devices.



Mono, stereo and surround formats

Channel layout determines how many audio streams and spatial cues must be preserved.



MPEG audio standards: early perceptual codecs

MPEG specifies the bitstream and decoder behavior, leaving room for encoder innovation.

- MPEG is the acronym for Moving Pictures Experts Group.
- MPEG standardizes the type of information that an encoder has to produce as well as the way in which the decoder has to decompress the information.

Standard	Standardization details	Bit rates (kb/s)	Sampling rates (kHz)	Channels	Related references	Related information
MPEG-1	ISO/IEC 11172-3 1992	32–448 (Layer I) 32–384 (Layer II) 32–320 (Layer III)	32, 44.1, 48	Mono (1), stereo (2)	[ISOI92] [Bran94a] [Shli94] [Pan95] [Noll93] [Noll95] [Noll97] [John99]	A generic compression standard that targets primarily multimedia storage and retrieval.
MPEG-2 BC/LSF	ISO/IEC 13818-3 1994	32–256 (Layer I) 8–160 (Layers II, III)	16, 22.05, 24	Multichannel Surround sound (5.1)	[ISOI94a] [Stol93a] [Gril94] [Noll93] [Noll95] [John99]	First digital television standard that enables lower frequencies and multichannel audio coding.
MPEG-2 NBC/AAC	ISO/IEC 13818-7 1997	8–160	8–96	Multichannel	[ISOI97b] [Bosi96] [Bosi97] [John99] [ISOI99] [Koen99] [Quac98b] [Gril99]	Advanced audio coding scheme that incorporates new coding methodologies (e.g., prediction, noise shaping, etc.).

MPEG audio standards: MPEG-4, MPEG-7, MPEG-21

Later standards broadened audio coding toward objects, metadata and multimedia frameworks.

Standard	Standardization details	Bit rates (kb/s)	Sampling rates (kHz)	Channels	Related references	Related information
MPEG-4 (Version 1)	ISO/IEC 14496-3 Oct, 1998	0.2–384	8–96	Multichannel	[Gril97] [Park97] [Edle99] [Purn00a] [Sche98a] [Vaan00] [Sche01]	The first content-based multimedia standard, allowing universality/interactivity and a combination of natural and synthetic material, coded in the form of objects.
MPEG-4 (Version 2)	ISO/IEC 14496-3/AMD-1, Dec, 1999	0.2–384 (finer levels of increment possible)	8–96	Multichannel	[ISOI00] [Kim01] [Herr00a] [Purn99b] [Alla99] [Hilp00] [Sper00] [Sper01]	
MPEG-7	ISO/IEC 15938-4 Sept, 2001	—	—	—	[ISOI01b] [Nack99a] [Nack99b] [Quac01] [Lind99] [Lind00] [Lind01] [Speng01] [ISOI02a]	A normative metadata standard that provides a Multimedia Content Description Interface.
MPEG-21	ISO/IEC-21000	—	—	—	[ISOI03a] [Borm03] [Burn03]	A multimedia framework that provides interoperability in content-access and distribution.

MPEG-1 Audio

Lossy compression for mono and stereo signals.

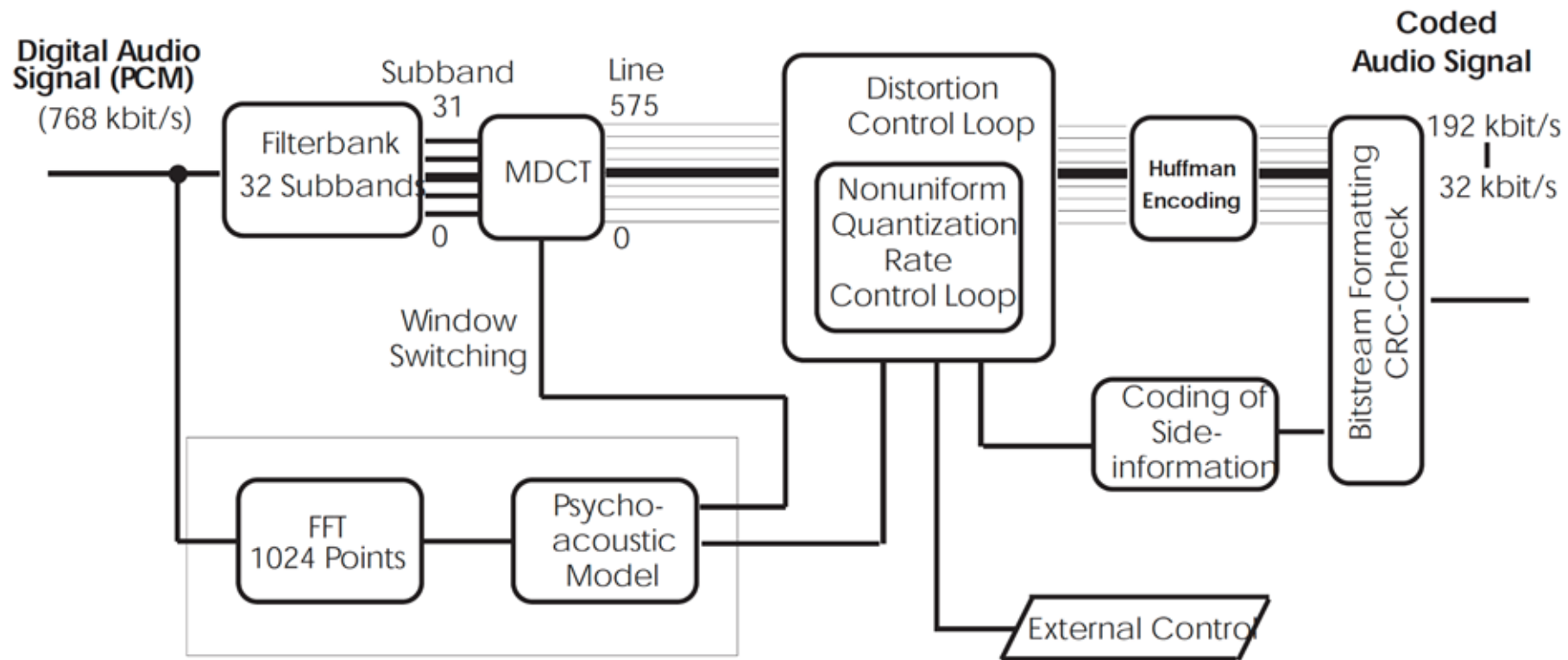
- Sampling frequencies: 32 kHz, 44.1 kHz and 48 kHz.
- Precision: 16 bits.
- Three compression layers:
 - Layer 1: 32–448 kb/s, target 192 kb/s
 - Layer 2: 32–384 kb/s, target 128 kb/s
 - Layer 3: 32–320 kb/s, target 64 kb/s
- Layer 3 is the MP3 format, widely used for Internet applications.

Layer 3 = MP3

**Same family, increasing
compression complexity
from Layer 1 to Layer 3.**

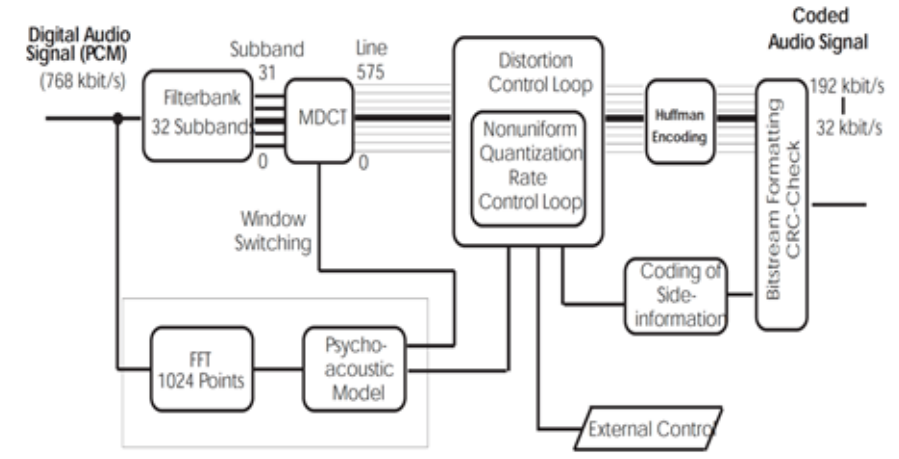
MP3 encoder

Hybrid filter bank, psychoacoustic model, nested rate/distortion loops and Huffman coding.



MP3 encoder: main components

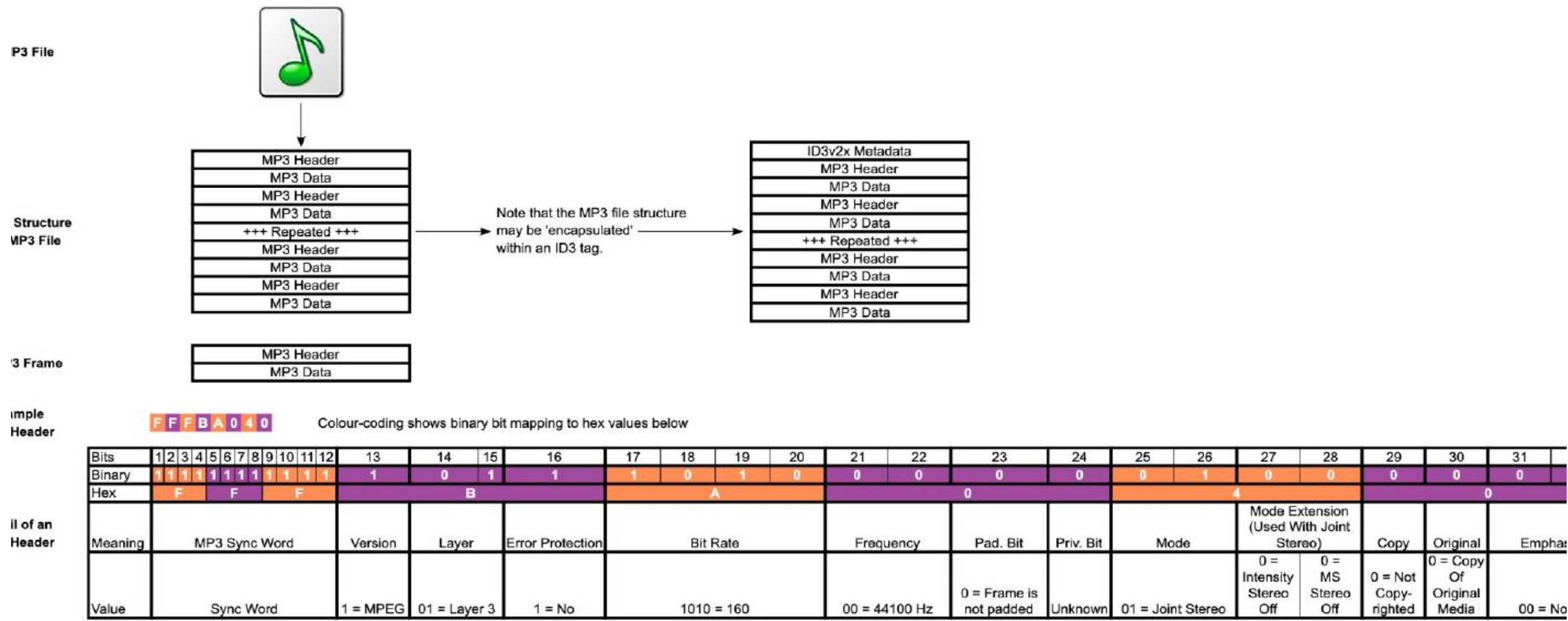
- Filter bank: hybrid filter bank — polyphase filter bank followed by MDCT.
- Perceptual model: estimates masking thresholds using FFT and critical bands.
- Quantization and coding: two nested iteration loops and Huffman coding.



The standard fixes the decoder/bitstream; encoder quality depends on model and rate-control decisions.

MP3 audio format and bitstream fields

An MP3 file is a sequence of frames, often with metadata and per-frame headers.



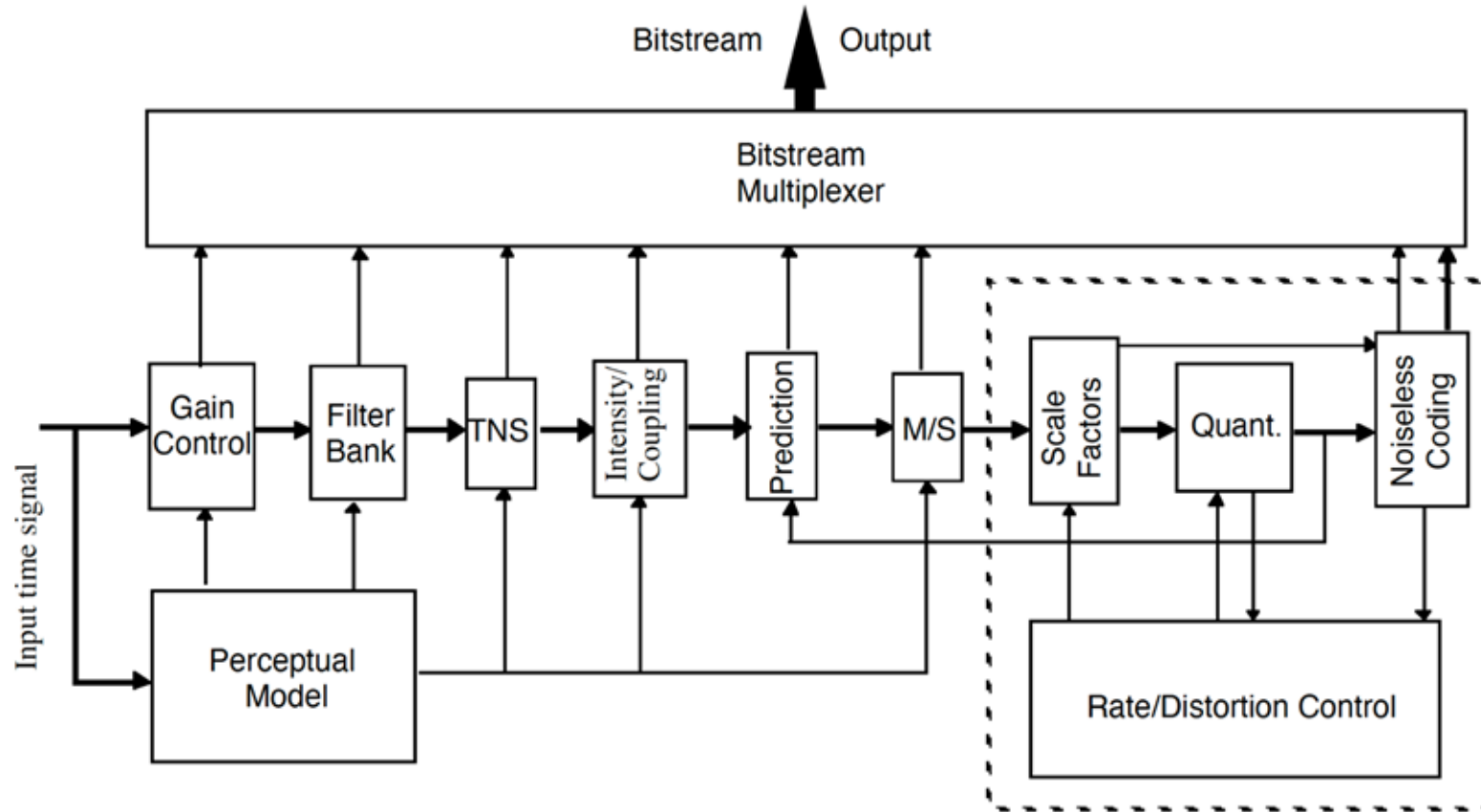
MPEG-2 Advanced Audio Coding (AAC)

AAC follows the Layer-3 paradigm but adds tools for improved low-bitrate quality.

- Supports up to 5.1 channels.
- AAC follows the same basic coding paradigm as Layer-3 (high frequency resolution filterbank, non-uniform quantization, Huffman coding, iteration loop structure using analysis-by-synthesis), but improves on Layer-3 in a lot of details and uses new coding tools for improved quality at low bitrates.
 - Higher frequency resolution (1024 frequency lines)
 - Improved joint stereo coding
 - Improved Huffman coding
 - Enhanced block switching

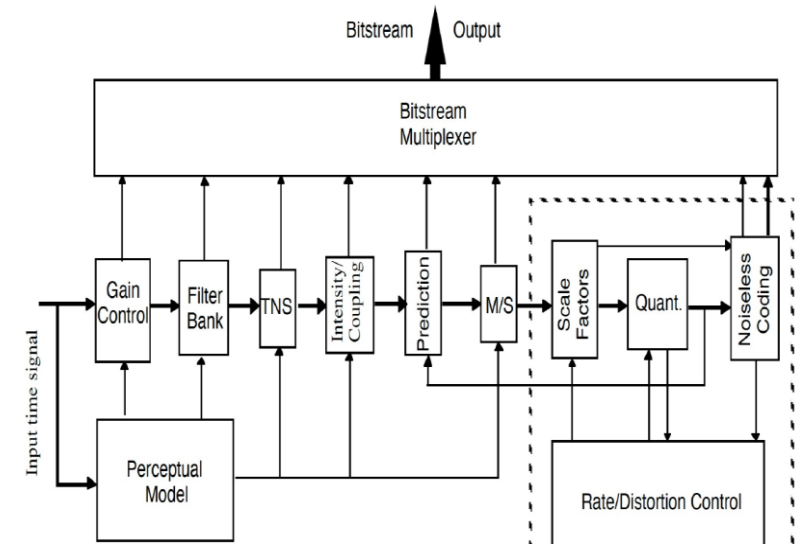
AAC encoding

AAC extends the perceptual transform-coding architecture with additional coding tools.



AAC coding tools

- Temporal Noise Shaping (TNS): controls the temporal shape of quantization noise.
- Intensity coding and coupling: reduce perceptually irrelevant high-frequency channel information.
- Prediction: removes redundancies between adjacent frames.
- M/S coding: removes stereo redundancy by coding sum and difference rather than left and right.



AAC improves MP3 mainly by adding better tools around the same core transform-coding idea.

MPEG-4 Audio

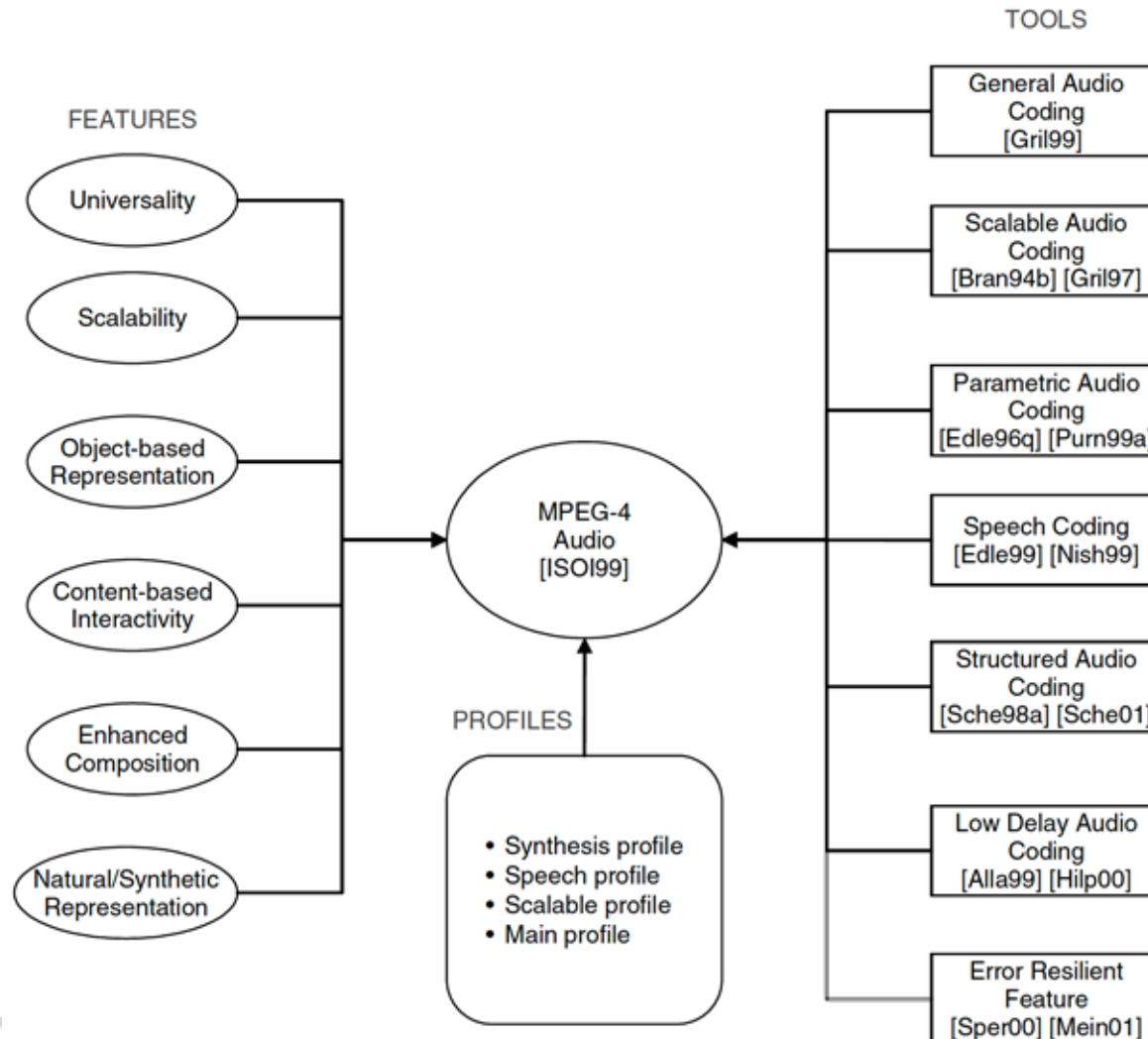
A broader framework with scalable, object-based and natural/synthetic audio representations.

- Extensive scalability and object-based representations.
- Variety of applications.
 - General audio signals
 - Speech signals
 - Synthetic audio
 - Synthesized speech (structured audio)
- More sample frequencies (8-96 kHz).
- Arbitrary bit rates and variable frame length.
- Higher efficiency and simpler filterbank.
 - Uses pure MDCT (modified discrete cosine transform)
 - Used in Windows Media Audio

MPEG-4 moves beyond one codec: it organizes a family of tools and profiles.

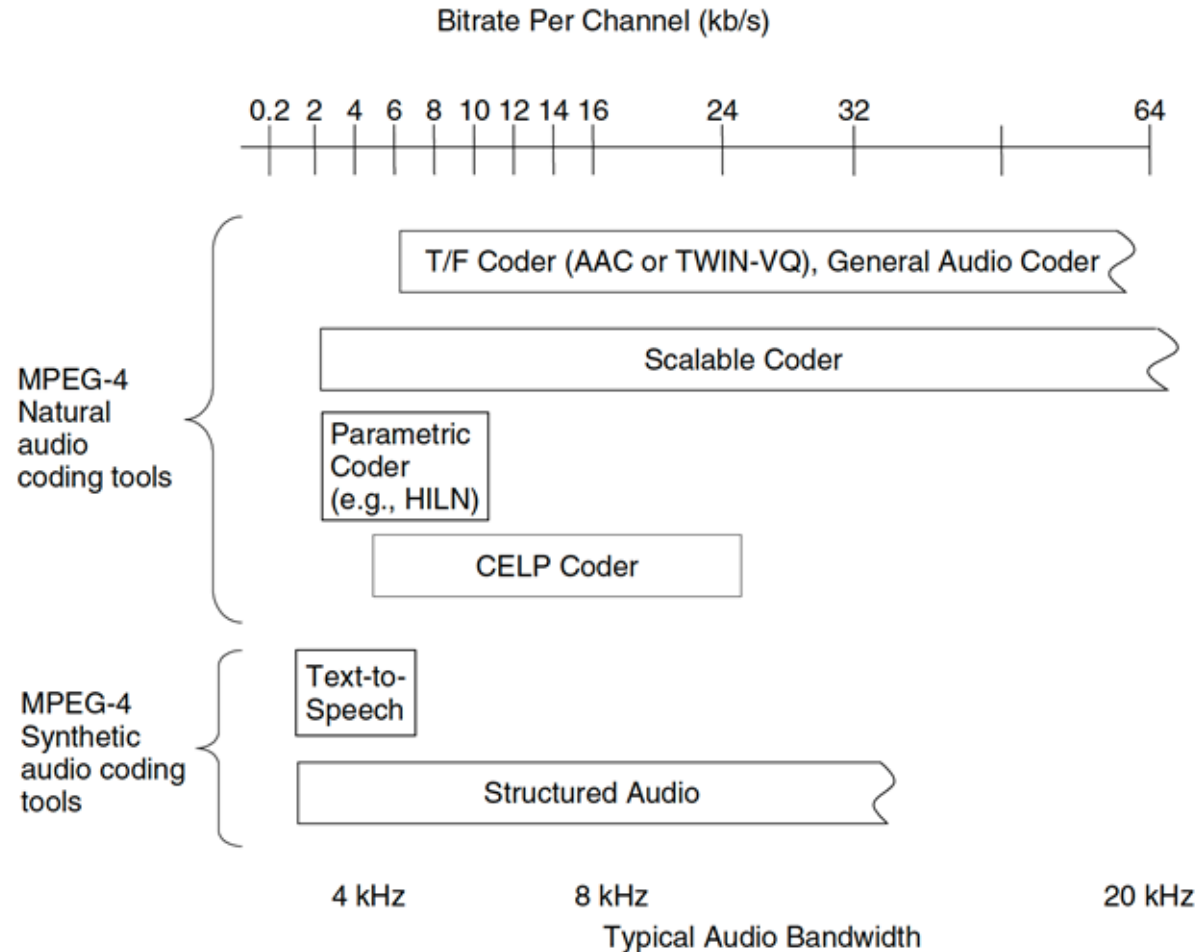
MPEG-4 Audio: features, profiles and tools

MPEG-4 integrates natural audio, speech, synthetic audio and error-resilient features.



MPEG-4 integrated tools

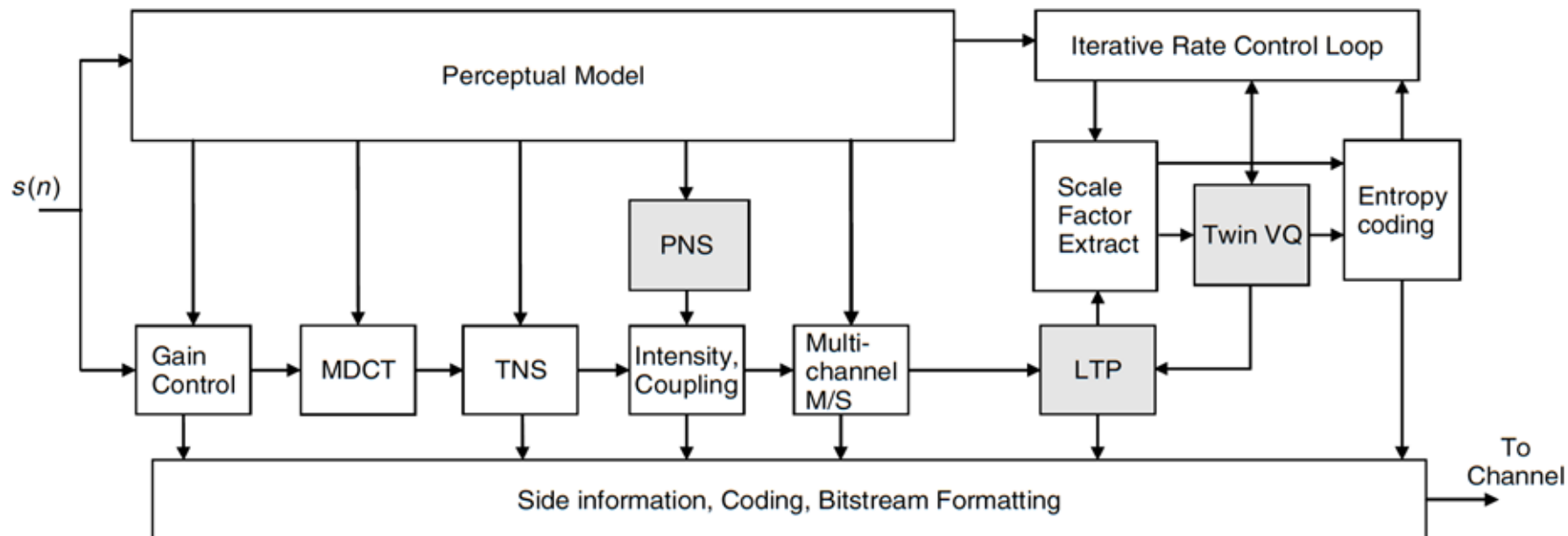
Different tools target different bit rates, bandwidths and signal types.



MPEG-4 General Audio coder

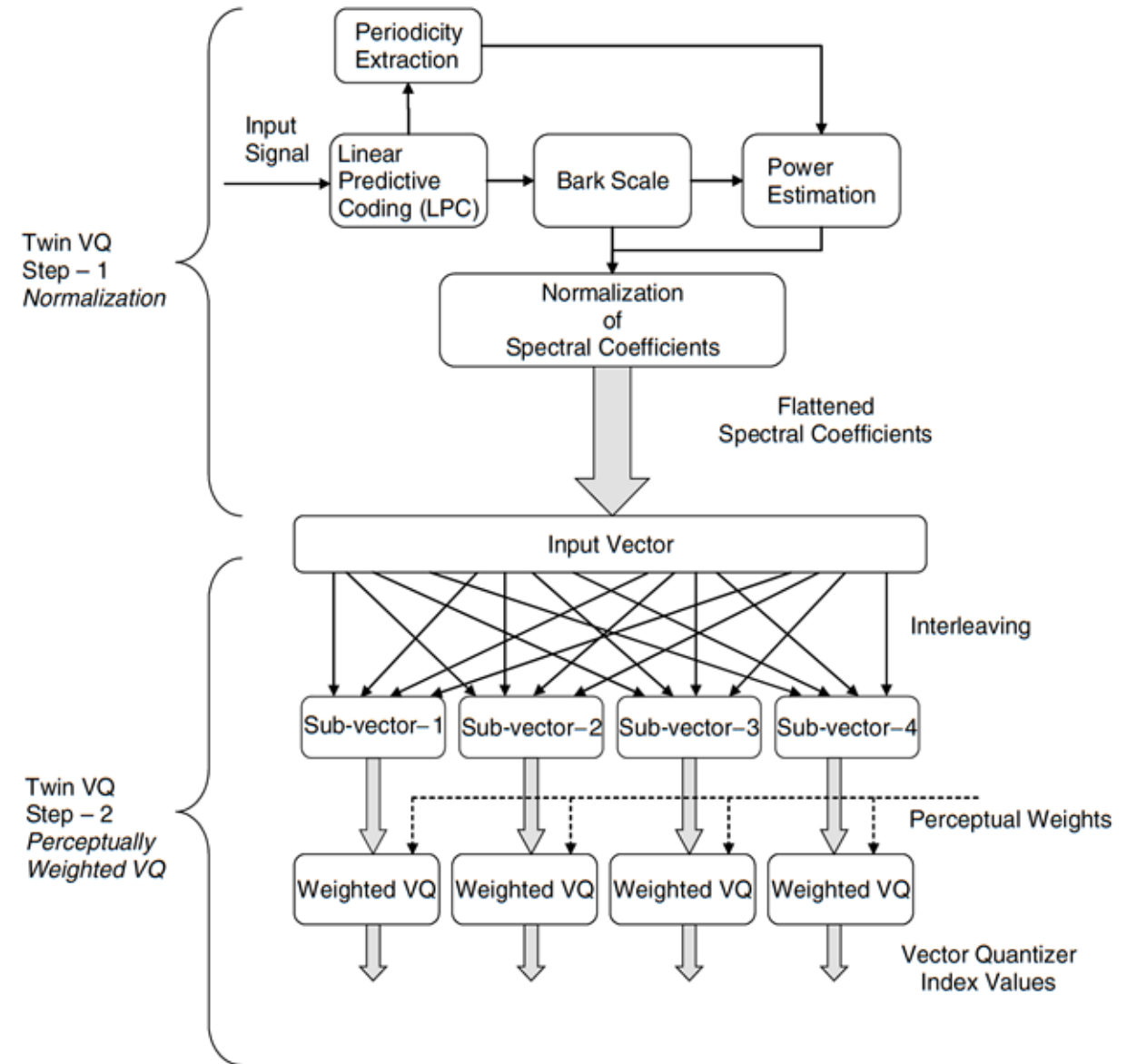
Built around MPEG-2 AAC with PNS, LTP, TwinVQ coding and scalability.

- Built around MPEG-2 AAC.
- Extended features and coder configurations given by the perceptual noise substitution (PNS), long term predictions (LTP), Twin VQ coding, and scalability.



MPEG-4 Twin VQ in GA coder

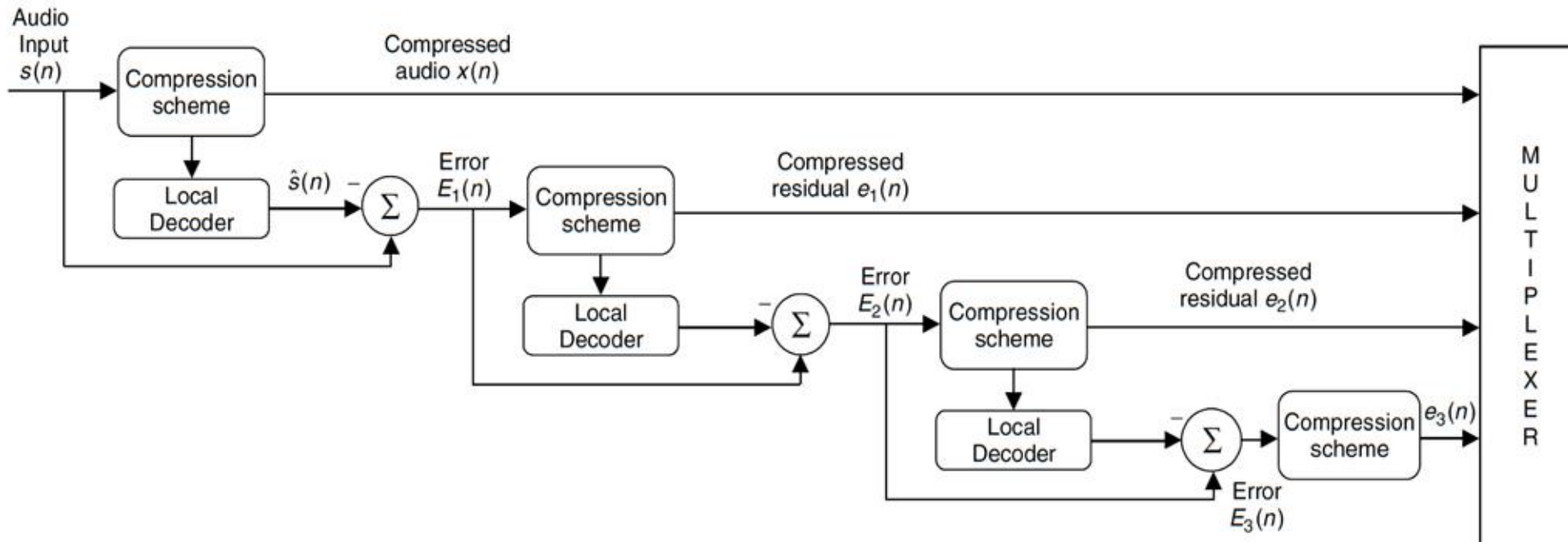
- Transform-domain Weighted Interleave Vector Quantization.
- Vector quantization of transformed spectral coefficients.
- Designed for low bit rates: 6–8 kb/s.



MPEG-4 scalable audio coding

Scalable coding supports variable-rate decoding depending on channel capacity.

- Variable rate encoding/decoding depending on transmission channel capacity.
- Decoding at low bit rates (16, 32, or 64 kb/s within high-rate code (64 kb/s).



MPEG-4 speech coding

Speech tools include HVXC and CELP with bit rates from 2 to 24 kb/s.

- Bit rates between 2 kb/s and 24 kb/s.
- Based on harmonic vector excitation coding (HVXC) and code excited predictive coding (CELP).

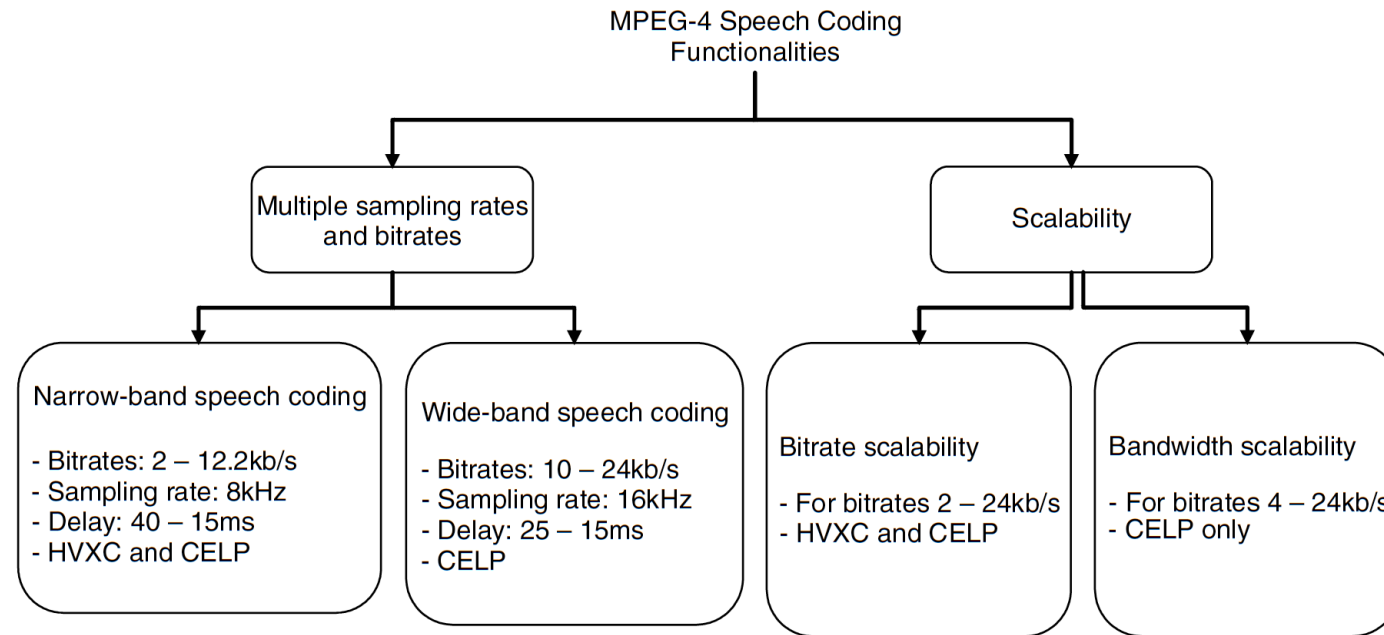


Figure 10.17. MPEG-4 speech coder functionalities.

MPEG-4 structured audio coding

Structured Audio represents synthetic audio through score, orchestra and synthesis descriptions.

- Ability to represent and encode efficiently synthetic audio and multimedia content.
- Based on the synthesizer-description language called *Csound*.
- Defines a set of syntax and semantic rules corresponding to the Structured Audio orchestra Language (SAOL).

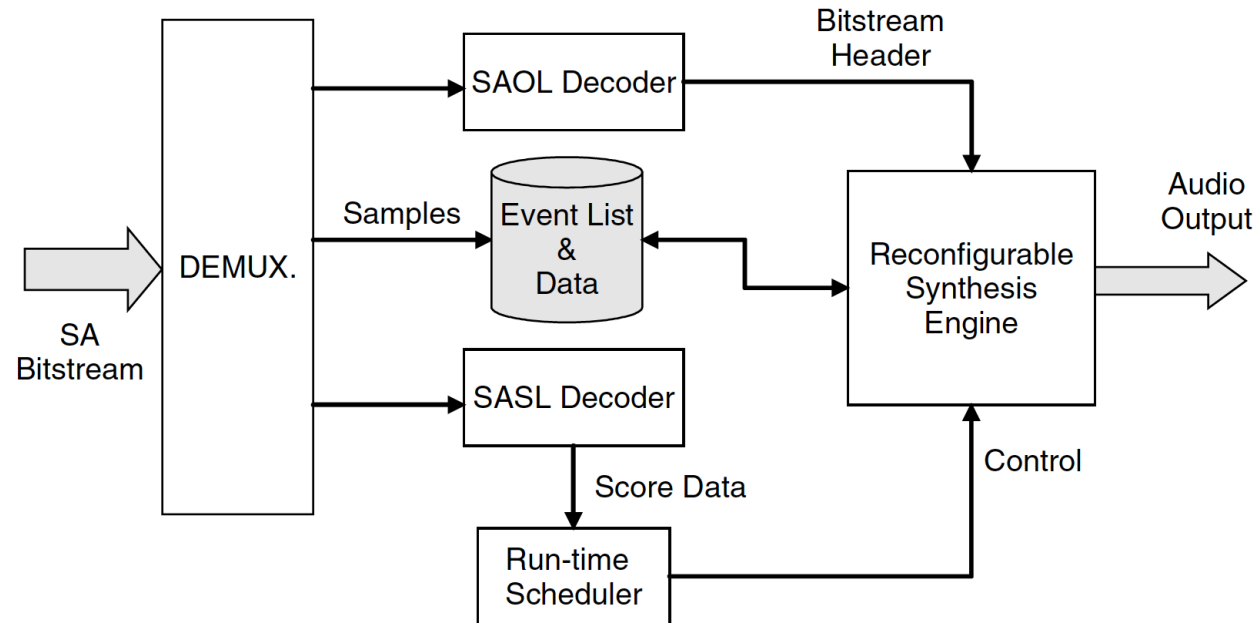


Figure 10.18. MPEG-4 SA decoder (after [Sche98a]).

Dolby AC-3 (Dolby Digital)

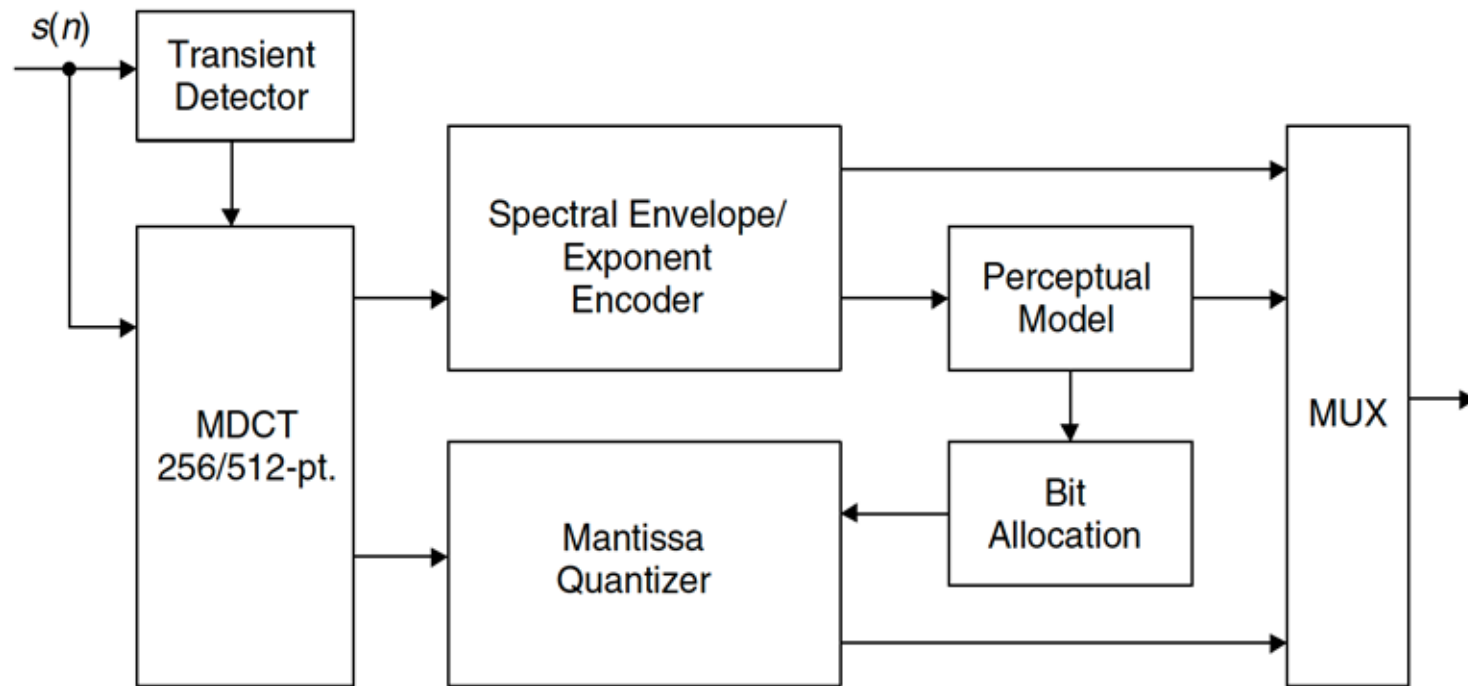
A de facto standard for movie theaters and home theater systems.

- Sample rates: 32, 44.1, and 48 kHz
- Bit rates: 32–640 kb/s, variable
- High-quality output at 64 kb/s per channel
- Delay roughly 100 ms
- MCDT filter bank
- Multiple channels processed as an ensemble
- Robust decoder downmix functionality from 5.1 to fewer channels

AC-3 was designed around multichannel playback and robust consumer decoding.

Dolby Digital encoder

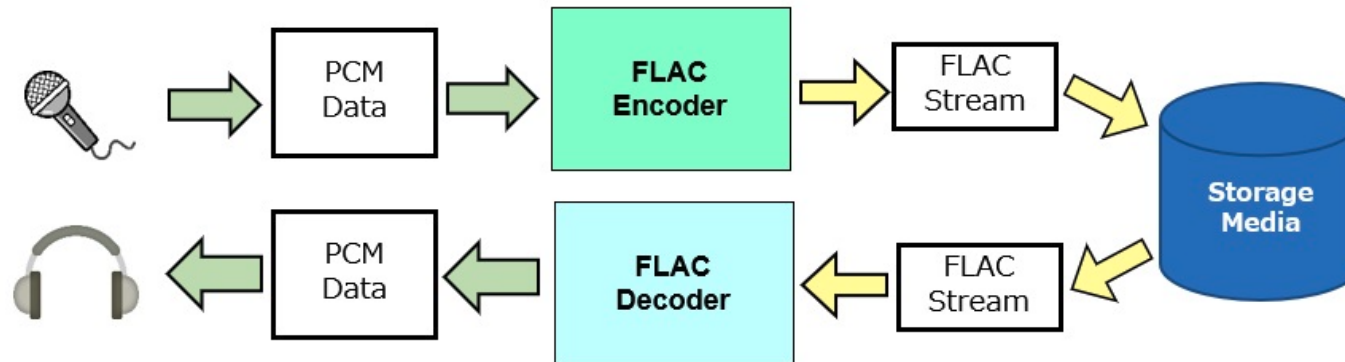
MDCT, transient detection, perceptual model, bit allocation and mantissa quantization are combined before multiplexing.



Lossless audio coding: FLAC

FLAC preserves the original PCM audio exactly while reducing file size.

- High-quality audio without any artifacts.
- Compression ratios of 2:1 – 4:1.
- Similar to how Zip works, except with FLAC you will get much better compression because it is designed specifically for audio.
- Non-proprietary and has an open-source reference implementation.



FLAC properties

- **Lossless:** The encoding of audio (PCM) data incurs no loss of information, and the decoded audio is bit-for-bit identical to what went into the encoder.
- **Fast:** FLAC is asymmetric in favor of decode speed. Decoding requires only integer arithmetic, and is much less compute-intensive than for most perceptual codecs.
- **Hardware support:** FLAC is supported by dozens of consumer electronic devices, from portable players, to home stereo equipment, to car stereo.
- **Flexible metadata:** FLAC's metadata system supports tags, cover art, seek tables, and cue sheets. Applications can write their own APPLICATION metadata once they register an ID.
- **Seekable:** FLAC supports fast sample-accurate seeking. Not only is this useful for playback, it makes FLAC files suitable for use in editing applications.
- **Streamable:** Each FLAC frame contains enough data to decode that frame.

FFmpeg

A practical command-line tool and library suite for multimedia processing.

- Free and open-source project for handling video, audio and multimedia streams.
- Core tool: ffmpeg, widely used for processing audio and video files.
- Common uses: format transcoding, trimming and concatenation, scaling and post-production effects.
- Useful for standards-compliance workflows and reproducible audio-coding experiments.

In class, FFmpeg connects standards to actual files: inspect, transcode, compare bitrates and listen.

Key takeaways

- Audio standards define interoperable bitstreams and decoders; encoders can still differ greatly in quality.
- MP3, AAC and AC-3 combine filter banks/MDCT, psychoacoustics, quantization, entropy coding and bitstream formatting.
- AAC and MPEG-4 add more flexible tools for low bitrates, multichannel audio, scalability and object/synthetic representations.
- Lossless codecs such as FLAC preserve PCM exactly, but achieve lower compression ratios than perceptual codecs.
- Practical tools such as FFmpeg make standards observable through real files, metadata, codecs and bitrates.

References and further reading

- Andreas Spanias, Ted Painter, Venkatraman Atti (2007). Audio Signal Processing and Coding. Chapter 10: Audio coding standards and algorithms.
- Free Lossless Audio Codec (FLAC): xiph.org/flac/
- FFmpeg: ffmpeg.org/